

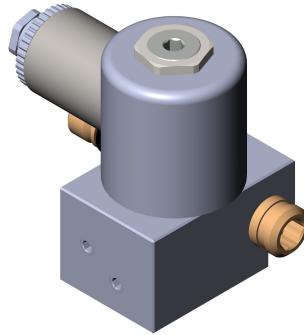


Instructions manual - UCM Valve iL10-S

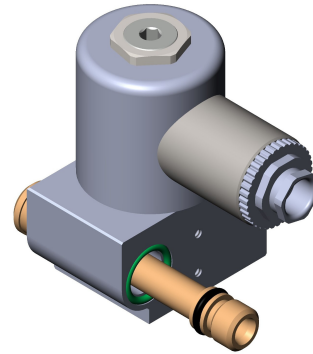
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**NEW
PRODUCT**



iL10-S front view



iL10-S back view



Warning: Only qualified personnel should adjust or service valves. Unauthorised manipulation may result in injury, loss of life or damage to equipment. Prior to servicing internal parts, ensure that the electrical power is switched off, ball valve is closed and residual pressure in the valve is reduced to zero. Very high-pressure spikes could result in deformation and oil splashing, which could cause serious injuries.

iL10-S Description: The **iL10-S** valves are solenoid operated lock valves designed as a safety option against Unintended Car Movement (UCM) for hydraulic elevators. The **iL10-S** model possesses only the adjustment 6 (Adj.6). Down movement of the lift is only possible by energizing coil D on the EV control valve and coil **A3** on the **iL10-S** valves. The upward direction is not affected.

Existing Blain valves of the EV, EV4 and EV40 assemblies in sizes $\frac{3}{4}$ ", 1", 1.5", 2" and 2.5" can easily be retrofitted with UCM safety using the **iL10-S** valve. The **iL10-S** model has a special **adjustment 6**. The type of adjustment 6 varies depending on the size of the control valve. Specify the actual valve size when ordering the **iL10-S** for renovation purposes. (See the table on page 4.)

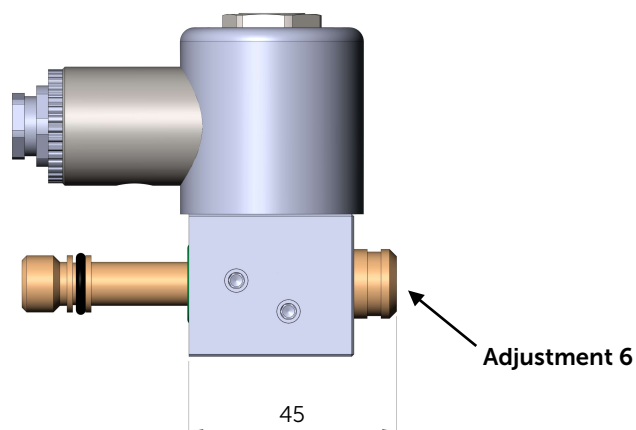
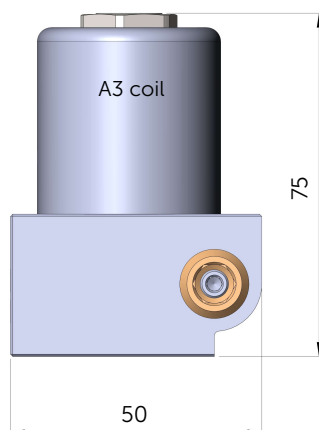
Technical data	unit	$\frac{3}{4}$ " iL10-S	1" - 2" iL10-S	2.5" iL10-S
Flow range:	l/min	125	800	1530
Pressure range:	bar	10-95	8-75	8-75
Burstpressure:	bar	p > 500	p > 500	340
Oil viscosity range:	cSt	~20 to 300 (~55°C to 10°C for ISO VG46)		
Oil temperature:	°C	10°C to 55°C		
Weight:	kg	0.6		

Coil voltages:

Alternating current (IP68) 50/60 Hz 24V/1.8A; 42V/1A; 110V/0.43A; 230V/0.18A

Direct current (IP 68) 12V/2A; 24V/1.1A; 42V/0.5A; 48V/0.6A; 80V/0.3A; 110V/0.25A; 196V/0.14A

Dimensions:



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Designer and Manufacturer of the highest quality control valves & safety components for hydraulic elevators

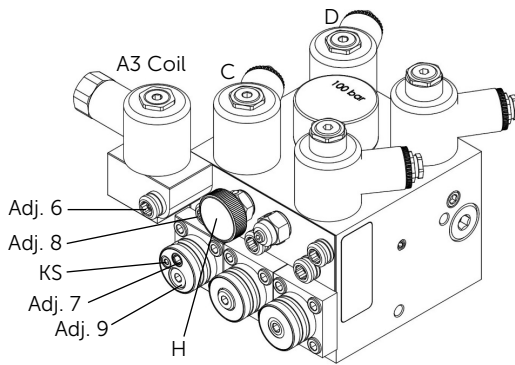


Figure 1: iL10-S valve assembly on 3/4" EV valve

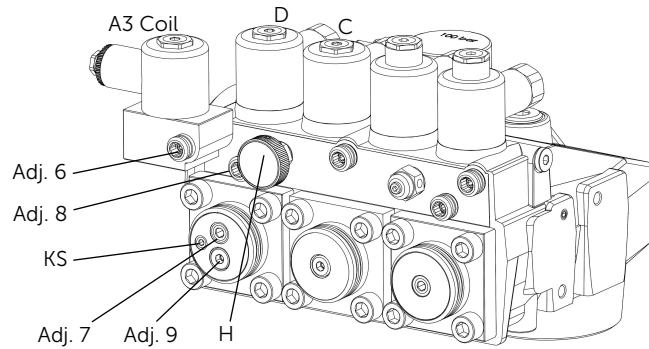


Figure 2: iL10-S valve assembly on 1.5"/2" EV valve

Adjustments

6	Adjustment-down acceleration	A3	iL10-S Solenoid coil
7	Adjustment-down speed	H	Manual lowering
8	Adjustment-down deceleration	KS	Slack rope valve (option)
9	Adjustment-down levelling		

iL10-S UCM Action

Rest Position: When **iL10-S** is in stand-by, the solenoid **A3** de-energized and the down valve **X** (inside the control valve) is kept closed by the cylinder pressure and a spring force preventing flow from cylinder to the tank.

Up Travel: During up travel with the pump running, oil flows through the check valve **V** and out through the port **Z** to the main cylinder. The coils of solenoids **A3**, **C** and **D** are not energized.

Down Travel: In order to have a down travel, the **A3** coil of **iL10-S** should be energized approximately 0.5s earlier than the coils on the main control valve (e.g. **D** & **C** of the EV100). When **A3** and the down coils **D** & **C** are energized the down valve **X** opens allowing the oil flow from cylinder to the tank (from **Z** to **T**), that is, lift starts a down travel. As the elevator reaches the stop switch at slow speed (**C** coil is de-energized), first the down coil **D** of main control valve is de-energized. About 0.5s later, the coil **A3** is de-energized on the **iL10-S** valve. In this way, it is made sure that the down valve **X** is at its closed position when the coil **A3** is switched off.

Pressure loss: The **iL10-S** valves does not cause extra pressure loss in the main hydraulic line.

Emergency down: To lower the cabin in an emergency situation, the self-closing emergency lowering valve **H** on the control valve is operated. When it is opened, the control oil behind the down flow guide escapes into the tank and the down flow guide opens. The lowering speed is determined by setting **9**.

The **KS** option prevents a slack rope condition in a 2:1 roped elevator. If the cabin rests on the buffer after manual lowering with **H**, the **KS** option prevents the piston rod from sinking further and thus the slack rope condition.

Adjustments

Control valve adjustment: **iL10-S** valves function effectively when the down travel adjustments on the main control valve are done acceptably. Too long, unacceptable deceleration time may increase stopping distance. The main control valve is adjusted similar to the EV type valves. When solenoids **C**, **D** and **A3** are energised, the car will accelerate downwards according to the setting of the adjustments:

6. Down Acceleration: 'In' (clockwise) provides a softer down acceleration, 'out' (c-clockwise) a quicker acceleration.

7. Down Speed: With solenoids **C**, **D** and **A3** energised, the full down speed of the car is according to the setting of adjustment **7**. 'In' (clockwise) provides a slower down speed, 'out' (c-clockwise) a faster down speed.

8. Down Deceleration: When solenoid **C** is de-energised whilst solenoid **D** and **A3** remain energised, the car will decelerate into slow speed according to the setting of adjustment **8**. 'In' (clockwise) provides a softer deceleration, 'out' (c-clockwise) a quicker deceleration.

Attention: Do not close all the way in! Closing Adjustment **8** completely (clockwise) may cause the car to fall on the buffers. If setting **8** is changed significantly, settings **6** and **9** must be readjusted.

9. Down Levelling: If only coils **D** and **A3** energised, the car will proceed at its down levelling speed according to the setting of adjustment **9**. 'In' (clockwise) provides a slower, 'out' (c-clockwise) a faster down levelling speed.

Down Stop: When solenoid coil **D** is also de-energised (coil **C** is already de-energised), the car will stop according to the setting of Adjustment **8**. Coil **A3** should be de-energized after a delay of 0.5s.

Manual Down Speed: When the Manual Lowering Knob **H** is opened, the elevator travels down with levelling speed according to the Adjustment **9**.

Slack Rope Valve KS: With **KS** turned all the way 'in', then half a turn back out, the unloaded car should descend when Manual Lowering **H** is opened. Should the car not descend, **KS** must be backed off (CCW) until the car just begins to descend, then backed off a further turn to ensure that with cold oil, the car can be lowered as required. If the cabin is sitting on the buffer, the piston must not move any further downwards when **H** is opened. The **KS** setting can be adjusted by turning the **KS** screw in (cw) to increase the pressure, and out (ccw) to decrease the pressure.

Solenoid seat: Depending on the valve size and maximum working pressure, seat diameter size is changed in the **iL10-S** valves. This is shown in Table on page 4.

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Maintenance

Regular maintenance of the **iL10-S** is not required; however, an annual inspection is recommended. If internal leakage is detected, the components **DS** and **DN** in solenoid valve **D** (EV control block) and in solenoid valve **A3** (iL10-S) should be checked first. Subsequently, the condition of the O-rings at **V**, **X**, and **H** on the EV control block should be examined.

During the inspection, the condition of the two **FD** filters in both solenoid valve **D** and solenoid valve **A3** should be checked and cleaned if necessary. The filter for the adjustment **8** should also be inspected and cleaned as needed.

Functional test

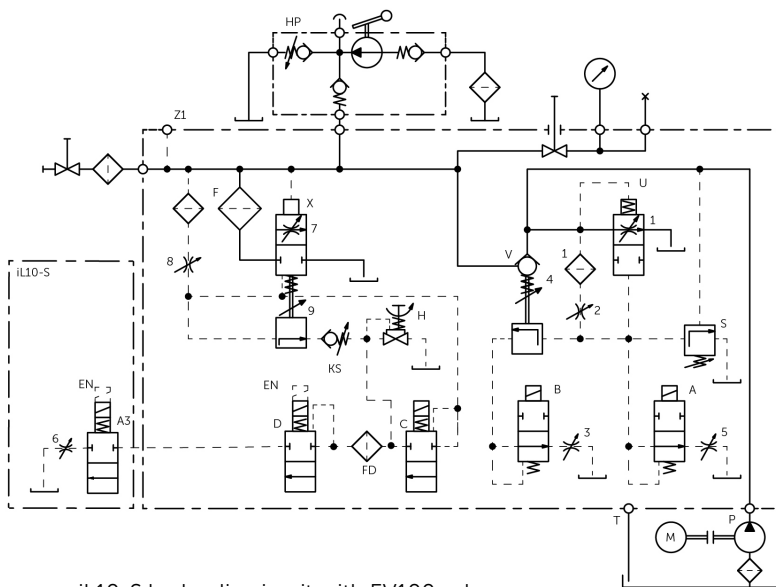
In order to check the functionality of the **iL10-S** valve, the coil **A3** can be de-energized during down travel. Alternatively, the **iL10-S** can also be tested by unscrewing the retaining nut **MM** and manually lifting the coil **A3**.

Caution! If the solenoid coil **M** is removed from the solenoid tube **DR** while under voltage for more than 10 seconds, this may cause deformation of the coil body or burning out of the solenoid coil **M**.

Firstly, the main control valve must be adjusted to ensure the elevator provides a smooth downward ride with acceptable deceleration times. To test the UCM (Unintended Car Movement) function, the **A3** coil of the **iL10-S** valve is deactivated during the descent of the elevator, or the **A3** coil is manually removed from the solenoid valve. The elevator should then stop within a travel distance of 800 mm.

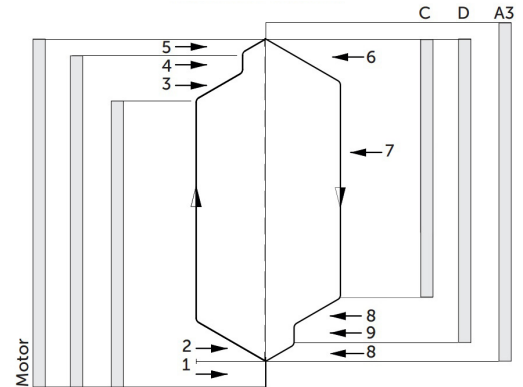
Status of the elevator	Power to coil A3	Power to coil D	Power to coil C
Up travel and relevelling	power off		
Down travel	power on		power on
Down relevelling			power off
Stop with closed doors	power off		
Emergency lowering	power on		power off
Emergency manual lowering	power off		
Hand pump operation	power off		

Hydraulic circuit

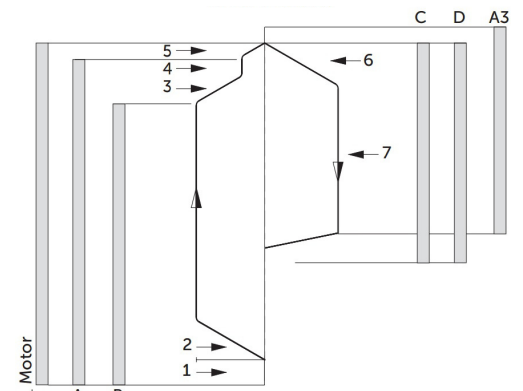


iL10-S hydraulic circuit with EV100 valve

Electrical sequence



normal up and down travel

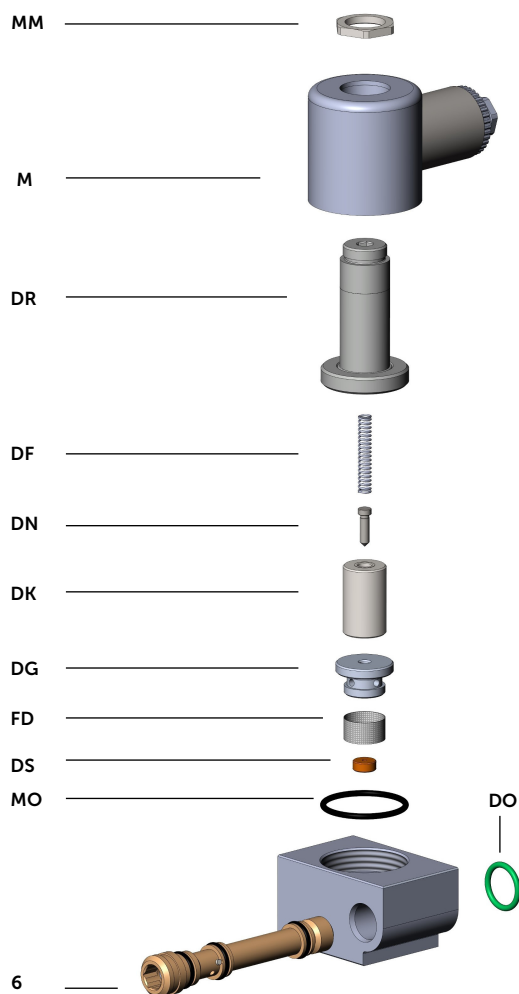


normal up travel, UCM case in down direction

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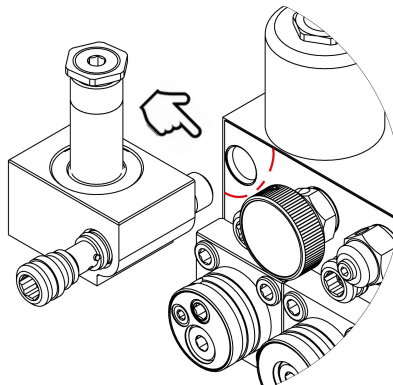
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iL10-S Explosion view

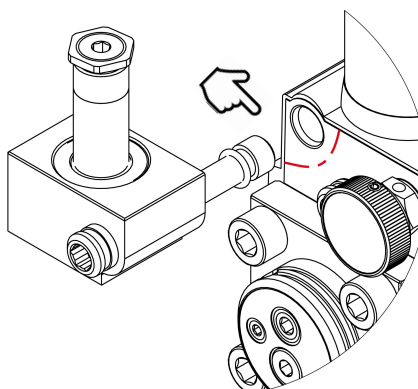


Caution!

After assembling the **iL10-S** on the EV block, if any external leakage is detected, remove the iL10-S block and cut away the aluminium front shield to ensure that no part of the shield remains between the iL10-S and EV valves. Then, clean the surface of the EV block and reassemble the iL10-S valve.



iL10-S with 3/4" EV100 valve



iL10-S with 1.5" EV100 valve

While ordering iL10-S

When placing an order, please specify the valve size (e.g., 2" EV100 or a pump flow rate of 250 l/min) and the working pressure range (e.g., 28bar to 55 bar). Depending on the pressure range and pump flow, different size of adjustment 6 and inserts are used. The seat discs (DS) vary depending on the operating pressure.

MM	Solenoid nut
M	Coil
DR	Solenoid tube
DF	Solenoid spring
DN	Needle
DK	Iron core
DG	Seat housing
FD	Filter solenoid
DS	Solenoid seat
MO	O-ring (26x2)
DO	O-ring (13x2)
6	Adjustment 6

Valve size	Maximal flow	SKU Adjustment 6
EV 3/4"	125 l/min	501177
EV 1.5"-2"	800l/min	501178
EV 2.5"	1500 l/min	501179

size	DS hole diameter	
	p_max ≤ 50 bar	p_max > 50 bar
EV 3/4" / EV 1.5"-2"	2.0 mm	1.6 mm
EV 2.5"	p_max ≤ 34 bar	p_max > 34 bar
	2.9 mm	2.5 mm

Select size of solenoid seat (DS) according to the maximum system pressure (cabin with payload).